Problem 1: Preliminaries and Warm-up

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1.

Show that $R_{\mathbf{n}}(\theta) = e^{-i\frac{\theta}{2}\mathbf{n}\cdot\boldsymbol{\sigma}}$ takes the form

$$\cos(\theta/2)I - i\sin(\theta/2)(n_X X + n_Y Y + n_Z Z), \tag{1}$$

where $\mathbf{n}$ is a unit 3D vector and $\boldsymbol{\sigma} = (X, Y, Z)$.

Solution: 首先计算 $(\mathbf{n}\cdot\boldsymbol{\sigma})^2$。Pauli 矩阵满足对易关系 $\{X,Y\} = \{Y,Z\} = \{Z,X\} = 0$ 和 $X^2 = Y^2 = Z^2 = I$，因此

$$\begin{aligned}
(\mathbf{n}\cdot\boldsymbol{\sigma})^2 &= (n_X X + n_Y Y + n_Z Z)^2 \\
&= n_X^2 X^2 + n_Y^2 Y^2 + n_Z^2 Z^2 + n_X n_Y (XY + YX) + n_Y n_Z (YZ + ZY) + n_Z n_X (ZX + XZ) \\
&= (n_X^2 + n_Y^2 + n_Z^2) I \\
&= I,
\end{aligned}$$

最后一步利用了 $\mathbf{n}$ 是单位向量。

基于此，将指数函数作 Taylor 展开并按奇偶项分组：

$$\begin{aligned}
R_{\mathbf{n}}(\theta) = e^{-i\frac{\theta}{2}\mathbf{n}\cdot\boldsymbol{\sigma}} &= \sum_{k=0}^{\infty} \frac{1}{k!}\left(-i\frac{\theta}{2}\right)^k (\mathbf{n}\cdot\boldsymbol{\sigma})^k \\
&= \sum_{m=0}^{\infty} \frac{(-i\frac{\theta}{2})^{2m}}{(2m)!}(\mathbf{n}\cdot\boldsymbol{\sigma})^{2m} + \sum_{m=0}^{\infty} \frac{(-i\frac{\theta}{2})^{2m+1}}{(2m+1)!}(\mathbf{n}\cdot\boldsymbol{\sigma})^{2m+1}.
\end{aligned}$$

对于偶数项：$(\mathbf{n}\cdot\boldsymbol{\sigma})^{2m} = [(\mathbf{n}\cdot\boldsymbol{\sigma})^2]^m = I^m = I$；对于奇数项：$(\mathbf{n}\cdot\boldsymbol{\sigma})^{2m+1} = (\mathbf{n}\cdot\boldsymbol{\sigma})^{2m}(\mathbf{n}\cdot\boldsymbol{\sigma}) = \mathbf{n}\cdot\boldsymbol{\sigma}$。

代入：

$$\begin{aligned}
R_{\mathbf{n}}(\theta) &= \sum_{m=0}^{\infty} \frac{(-1)^m(\theta/2)^{2m}}{(2m)!} I - i\,(\mathbf{n}\cdot\boldsymbol{\sigma}) \sum_{m=0}^{\infty} \frac{(-1)^m(\theta/2)^{2m+1}}{(2m+1)!} \\
&= \cos(\theta/2)\, I \;-\; i\sin(\theta/2)\,(\mathbf{n}\cdot\boldsymbol{\sigma}) \\
&= \cos(\theta/2)I - i\sin(\theta/2)(n_X X + n_Y Y + n_Z Z).
\end{aligned}$$

这里我们利用了 $(-i)^{2m} = (-1)^m$ 和 $(-i)^{2m+1} = -i(-1)^m$。

□

2.

Show that any single-qubit unitary U takes the form $e^{i\alpha}R_{\mathbf{n}}(\theta)$ for some α, $\mathbf{n}$, and θ.

Solution: 任意 2×2 酉矩阵 U 满足 $U^\dagger U = I$ 且 $|\det U| = 1$。设 $\det U = e^{i\gamma}$，则可提取全局相位：

$$U = e^{i\gamma/2}\,S, \qquad S \in \mathrm{SU}(2), \quad \det S = 1.$$

令 $\alpha = \gamma/2$。下面只需证明任意 SU(2) 矩阵 S 均可写为 $R_{\mathbf{n}}(\theta)$ 的形式。

SU(2) 矩阵的最一般形式为

$$S = \begin{pmatrix} a & -b^* \\ b & a^* \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

令 $a = a_r + ia_i$，$b = b_r + ib_i$，则约束为 $a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$。

另一方面，由第 1 问的结果：

$$R_{\mathbf{n}}(\theta) = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}(n_X X + n_Y Y + n_Z Z) = \begin{pmatrix} \cos\frac{\theta}{2} - in_Z\sin\frac{\theta}{2} & -i(n_X - in_Y)\sin\frac{\theta}{2} \\ -i(n_X + in_Y)\sin\frac{\theta}{2} & \cos\frac{\theta}{2} + in_Z\sin\frac{\theta}{2} \end{pmatrix}.$$

对比 S 的形式，取

$$a = \cos\frac{\theta}{2} - in_Z\sin\frac{\theta}{2}, \qquad b = -i(n_X + in_Y)\sin\frac{\theta}{2},$$

则有 $a_r = \cos\frac{\theta}{2}$, $a_i = -n_Z\sin\frac{\theta}{2}$, $b_r = n_Y\sin\frac{\theta}{2}$, $b_i = -n_X\sin\frac{\theta}{2}$。由 $|\mathbf{n}| = 1$ 可知 $|a|^2 + |b|^2 = \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2} = 1$，即该映射覆盖了所有 SU(2) 矩阵。

具体地，给定额外的约束 $S^\dagger S = I$ 及 $\det S = 1$，可以直接解出参数：

$$\theta = 2\arccos(\mathrm{Re}(a)) = 2\arccos(a_r),$$
$$n_X = -\frac{\mathrm{Im}(b)}{\sin(\theta/2)}, \quad n_Y = \frac{\mathrm{Re}(b)}{\sin(\theta/2)}, \quad n_Z = -\frac{\mathrm{Im}(a)}{\sin(\theta/2)}.$$

由于 $a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$，$n_X^2 + n_Y^2 + n_Z^2 = 1$ 自然满足。

因此任意单量子比特酉变换 U 都可写为 $U = e^{i\alpha}R_{\mathbf{n}}(\theta)$。

□

3.

Implement $R_{\mathbf{n}}(\theta)$ using $R_Y(\alpha)$ and $R_Z(\beta)$ for appropriate choices of α and β.

Solution: 采用 Euler 角分解。将单位向量 $\mathbf{n}$ 写成球坐标形式：

$$\mathbf{n} = (\sin\beta\cos\alpha,\ \sin\beta\sin\alpha,\ \cos\beta),$$

其中 $\alpha \in [0, 2\pi)$，$\beta \in [0, \pi]$。

此时

$$\mathbf{n}\cdot\boldsymbol{\sigma} = \sin\beta\cos\alpha\,X + \sin\beta\sin\alpha\,Y + \cos\beta\,Z.$$

利用 Pauli 矩阵的相似变换：

$$e^{-i\frac{\alpha}{2}Z}Xe^{i\frac{\alpha}{2}Z} = \cos\alpha\,X + \sin\alpha\,Y,$$
$$e^{-i\frac{\beta}{2}Y}Ze^{i\frac{\beta}{2}Y} = \cos\beta\,Z + \sin\beta\,X,$$

可以构造

$$\begin{aligned} e^{-i\frac{\alpha}{2}Z}e^{-i\frac{\beta}{2}Y}\,Z\,e^{i\frac{\beta}{2}Y}e^{i\frac{\alpha}{2}Z} &= e^{-i\frac{\alpha}{2}Z}(\cos\beta\,Z + \sin\beta\,X)e^{i\frac{\alpha}{2}Z} \\ &= \cos\beta\,Z + \sin\beta\cos\alpha\,X + \sin\beta\sin\alpha\,Y \\ &= \mathbf{n}\cdot\boldsymbol{\sigma}. \end{aligned}$$

即 $\mathbf{n}\cdot\boldsymbol{\sigma} = R_Z(\alpha)R_Y(\beta)\,Z\,R_Y(-\beta)R_Z(-\alpha)$，其中

$$R_Y(\phi) = e^{-i\frac{\phi}{2}Y}, \qquad R_Z(\phi) = e^{-i\frac{\phi}{2}Z}.$$

对任意解析函数 f，有 $f(R\,A\,R^\dagger) = R\,f(A)\,R^\dagger$。令 $f(x) = e^{-i\frac{\theta}{2}x}$，则

$$\begin{aligned} R_{\mathbf{n}}(\theta) = e^{-i\frac{\theta}{2}\mathbf{n}\cdot\boldsymbol{\sigma}} &= R_Z(\alpha)R_Y(\beta)\ e^{-i\frac{\theta}{2}Z}\ R_Y(-\beta)R_Z(-\alpha) \\ &= R_Z(\alpha)\,R_Y(\beta)\,R_Z(\theta)\,R_Y(-\beta)\,R_Z(-\alpha). \end{aligned}$$

这就是最一般的分解式。特别地，当 $\mathbf{n}$ 位于 XZ 平面时（即 $\alpha = 0$，$\mathbf{n} = (\sin\beta, 0, \cos\beta)$），退化为更简洁的形式：

$$R_{\mathbf{n}}(\theta) = R_Y(\beta)\,R_Z(\theta)\,R_Y(-\beta).$$

因此 $R_{\mathbf{n}}(\theta)$ 总可以通过 R_Y 和 R_Z 门的组合来实现，具体参数由 $\mathbf{n}$ 的球坐标 (α, β) 和旋转角 θ 共同确定。

$\square$

4. State Preparation

Starting with all qubits initialized in $|0\rangle$, prepare the following states using single-qubit gates, two-qubit gates, ancillary qubits, classical feedback, and measurements. The resource cost of the final circuit should be as small as possible. You may optimize for circuit size, circuit depth, or both simultaneously.

(a) Prepare an n-qubit quantum state $\frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle$;

(b) Prepare an n-qubit state $\alpha|000\cdots000\rangle + \sqrt{1-\alpha^2}|111\cdots111\rangle$ for any $\alpha \in \mathbb{R}$;

(c) Prepare the 4-qubit state $\frac{3}{10}|0001\rangle + \frac{2}{5}|0010\rangle + \frac{\sqrt{6}}{4}|0100\rangle + \frac{\sqrt{6}}{4}|1000\rangle$;

(d) (Optional) Prepare the following state:

$$\begin{aligned} &\alpha(|1000000\rangle + |0010101\rangle + |1110011\rangle + |0100110\rangle \\ &\quad + |1001111\rangle + |0011010\rangle + |1111100\rangle + |0101001\rangle) \\ &+ \beta(|0111111\rangle + |1101010\rangle + |0001100\rangle + |1011001\rangle \\ &\quad + |0110000\rangle + |1100101\rangle + |0000011\rangle + |1010110\rangle) \end{aligned}$$

(e) (Optional) Use TensorCircuit to verify your answers.

Solution:

(a) 对 n 个量子比特分别作用 Hadamard 门 H 即可，n 个单比特门，必然最优

(b) 先对第一个量子比特作用 $R_Y(\theta)$，其中 $\theta = 2\arccos\alpha$ ，使其变为 $\alpha|0\rangle + \sqrt{1-\alpha^2}|1\rangle$ ，然后以第一个量子比特为控制比特，对别的量子比特依次作用 CNOT 门，1 个单比特旋转门和 $n-1$ 个 CNOT 门

(c) 如果不用辅助比特的话应该只能一位一位做，并且需要 CCRy 和 CCCRy，消耗非常大，只要用一个辅助比特就可以比较优雅地完成这个制备，具体步骤如下：

(1) 对第 4 位作用 $R_Y(\theta_1)$ ，其中 $\theta_1 = 2\arcsin\frac{3}{10}$ ，变成 $\frac{3}{10}|0001\rangle + \frac{\sqrt{91}}{10}|0000\rangle$

(2) 以第 4 位为控制比特，当其为 0 时，对第 3 位作用 $R_Y(\theta_2)$ ，其中 $\theta_2 = 2\arcsin\frac{4}{\sqrt{91}}$ ，变成 $\frac{3}{10}|0001\rangle + \frac{2}{5}|0010\rangle + \frac{\sqrt{3}}{2}|0000\rangle$

(3) 对辅助比特 anc 作用 X 门，将其变为 $|1\rangle$ ，分别以第 3，4 位为控制比特，对 anc 作用 2 次 CNOT，然后以 anc 为控制比特，对第 2 位作用 $R_Y(\frac{\pi}{2})$ ，注意到此时状态为 $\frac{3}{10}|00010\rangle + \frac{2}{5}|00100\rangle + \frac{\sqrt{6}}{4}|01001\rangle + \frac{\sqrt{6}}{4}|00001\rangle$

(4) 以第 2 位为控制比特，对 anc 作用 CNOT，再以 anc 为控制比特，对第一位作用 CNOT，此时状态为 $\frac{3}{10}|00010\rangle + \frac{2}{5}|00100\rangle + \frac{\sqrt{6}}{4}|01000\rangle + \frac{\sqrt{6}}{4}|10001\rangle$

(5)（还原）以第 1 位为控制比特，对 anc 作用 CNOT，work done!

总共是 1 个辅助比特，5 个 CNOT，2 个 CRy，一个 X 和一个 Ry，相比一位一位做有显著提升

(d) (1) 先将后 4 位变成所有有偶数个 1 的态的均匀叠加，方法是：先对后 3 位作用 H 门，然后分别以后 3 位为控制比特，对第 4 位作用 CNOT

(2) 对第 1 位作用一个旋转门，将 $|0\rangle$ 变成 $2\sqrt{2}(\alpha|1\rangle + \beta|0\rangle)$ ，此时的态是

$$\begin{aligned}
&\alpha(|1000000\rangle + |1000101\rangle + |1000011\rangle + |1000110\rangle \\
&\quad + |1001111\rangle + |1001010\rangle + |1001100\rangle + |1001001\rangle) \\
&+ \beta(|0001111\rangle + |0001010\rangle + |0001100\rangle + |0001001\rangle \\
&\quad + |0000000\rangle + |0000101\rangle + |0000011\rangle + |0000110\rangle)
\end{aligned}$$

(3) 将前 3 位“校正”一下即可，具体地：

对第 1 位，用第 4，5 位做 CNOT

对第 2 位，作用 X 门之后，用第 1，5，6 位做 CNOT

对第 3 位，作用 X 门之后，用第 1，4，6 位做 CNOT

共计 11 个 CNOT，6 个单量子比特门，无辅助比特

□

5. Gate Construction

(a) Use CNOT gates to implement the SWAP operation: $\text{SWAP}|\psi\rangle|\phi\rangle = |\phi\rangle|\psi\rangle$.

(b) Consider four qubits q_0, q_1, q_2 and q_3. Implement two CNOT gates between q_0 and q_2, and q_1 and q_3 respectively, under the constraint that only CNOT gates between q_i and q_{i+1} are available. Try to use as few gates as possible.

(c) Use CNOT and R_Z gates to implement the two-qubit gate R_{ZZ}, where

$$R_Z(\theta) = e^{-i\theta/2Z}, \qquad R_{ZZ}(\theta) = e^{-i\theta/2Z\otimes Z}.$$

(d) Construct a Controlled-Phase gate

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\theta} \end{pmatrix}$$

using R_{ZZ} and single-qubit gates.

(e) For an arbitrary n-qubit Pauli operator P, construct a circuit to implement $e^{-i\theta P}$.

(f) Implement the following specific SU(4) gate:

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{2} & \frac{1}{\sqrt{2}} & 0 \\ \frac{\sqrt{3}}{2\sqrt{2}} & 0 & -\frac{\sqrt{3}}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2\sqrt{2}} & 0 & -\frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{2} \end{bmatrix}$$

using single-qubit pulses and CZ gates.

(g) (Optional) Use TensorCircuit to verify your answer above.

6. Measurement

(a) Construct a quantum circuit to measure the two-qubit operators Z_1Z_2 and X_1X_2 on $|\psi\rangle$ and project $|\psi\rangle$ onto the joint $+1$ eigenstate of Z_1Z_2 and X_1X_2.

(b) Give a general method to construct a quantum circuit to perform projective measurement of any Pauli operator P.

(c) Prepare the single-qubit quantum state $\alpha|0\rangle + \sqrt{1-\alpha^2}|1\rangle$ from an n-qubit quantum state $\alpha|000\cdots000\rangle + \sqrt{1-\alpha^2}|111\cdots111\rangle$ by measurement.

(d) (Optional) A 5-qubit fanout gate F_4 acting on qubits q_0, q_1, q_2, q_3, q_4 is defined as follows

$$F_4|x_0\rangle|x_1, x_2, x_3, x_4\rangle = |x_0\rangle|x_1 \oplus x_0, x_2 \oplus x_0, x_3 \oplus x_0, x_4 \oplus x_0\rangle,$$

where $x_0, x_1, \ldots, x_4 \in \{0, 1\}$. Two-qubit gates may act only on qubits (q_j, q_{j+1}) for $0 \leq j \leq 3$. Construct the quantum circuit of F_4 to minimize the circuit depth and size. (Hint: use ancillary qubits and measurements.)

(e) (Optional) Use CNOTs and Hadamards to compute the inner product $|\langle\psi|\phi|\rangle|$ for arbitrary states $|\psi\rangle$ and $|\phi\rangle$.

7. Reversible Circuit

(Optional) Construct a reversible classical circuit using Toffoli and NOT gates to evaluate the formula:

$$\begin{aligned} S_3(x) =&(\neg x_1 \vee \neg x_2) \wedge (x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_2 \vee \neg x_3) \\ &\wedge (\neg x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee x_3), \end{aligned}$$

where each x_i represents a Boolean value, and $\neg$, $\vee$, and $\wedge$ denote logical NOT, OR, and AND, respectively. Try to: (1) use as few ancillary qubits as possible, and (2) minimize the product of the number of gates and the total number of qubits.

8. Grover Search

(Optional) Using the results from the previous problem, find a configuration of x satisfying $S_3(x) =$ True with Grover's search algorithm. Implement your algorithm using TensorCircuit and verify the solution.